

THE FUTURE OF INTEGRATION: PROVE THE CONSEQUENCE

From trusted connection to locally authorized physical action

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

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This is a demonstration design, not a completed TA14 examination, certification, approval, or claim that any real building is safe. Every example, command, and disposition in this reference is hypothetical.

01 / Why this matters

Interoperability can move trusted information across systems. Semantic models can clarify its meaning. Analytics and autonomous agents can propose useful changes. None of these achievements, alone, establishes permission to actuate a particular asset at a particular time. This reference isolates the missing decision: the transition from proposed consequence to authorized execution.

Distinct propositions: Connection is not permission. Capability is not authority. Observed is not inferred. Inferred is not authorized. An admissible input does not create an admissible output.

02 / One bounded building scenario

In a hypothetical occupied commercial building, separate environmental and occupancy systems report elevated CO2 and temperature. A supervisory optimization agent proposes increasing outdoor-air intake and adjusting one air-handling unit for a 20-minute window. The proposal is not a direct command.

Scenario element	Frozen-test candidate
Target	AHU-1 serving occupied Zone A (illustrative identifiers)
Requested effect	Increase outdoor-air intake within an owner-approved envelope; no numerical setpoint assumed
Potential conflicts	Smoke-control status, equipment protection, outdoor conditions, demand response, delegation expiry
Required safe behavior	No actuation unless the local execution boundary admits this exact request

03 / The eight-stage TA14 chain

Stage	Question to prove	Minimum retained record
REALITY	What is happening at the asset now?	Current mode, relevant physical state, safety context
RECORD	What was actually observed and proposed?	Raw data, source IDs, timestamps, proposal ID
CONTINUITY	Can each relevant claim be traced without gaps?	Identity, time, mapping, delegation continuity
ADMISSIBILITY	Is the evidence fit for this decision?	Quality, uncertainty, conflicts, rejected claims
BINDING	Does authority apply to this actor, action and asset?	Scope, standing, validity, local restrictions
COMMIT	What bounded disposition is recorded?	Decision, rationale, limits, expiration, examiner
EXECUTION	Can the target enforce the authorized boundary?	Local permit/block evidence, interlocks, command trace
OUTCOME	What changed in physical reality?	Measured outcome, deviation, new evidence

04 / Authority must resolve locally

Authority context may cross organizational and system boundaries, including through derived delegation. But the target system must establish execution authority locally. A valid network session, trusted connection profile, or upstream operator role is not a substitute for checking the asset, action, timing, mode, and restrictions at the execution boundary.

05 / The four dispositions

Disposition	Illustrative trigger	Execution rule
ALLOW	Fresh admissible evidence; valid scope and standing; local constraints satisfied	Only the specifically bounded command may proceed
HOLD	Stale sensor data, unverified delegation, unknown operating mode	Block actuation pending resolution
DENY	Command conflicts with binding safety prohibition	Block and preserve adverse finding
ESCALATE	High-consequence ambiguity requiring qualified human judgment	Block until separate, valid authorization is established

A displayed HOLD is not enough. The test must demonstrate that the target did not execute the prohibited or unsupported command.

06 / Falsifiable examination matrix

Before testing, freeze the proposed action, asset identity, authority chain, rule set, observation window, expected safe state, and test harness. Each case below is a candidate test, not a finding.

Test ID	Injected condition	Expected boundary behavior
T01	Current evidence, valid delegation, no safety conflict	Bounded ALLOW; log actuation and outcome
T02	Expired or stale observation	HOLD; no command delivered
T03	Revoked owner/operator delegation	HOLD or DENY under frozen rule; no actuation
T04	Asset identifier or semantic mapping mismatch	HOLD; block target binding
T05	Smoke-control or equipment-protection conflict	DENY; preserve conflict and block
T06	Lost continuity between decision and actuation	Revalidate; block until new admissible decision
T07	Human override with no verified causal reach	No claimed effective intervention
T08	Changed operating mode after earlier ALLOW	Invalidate earlier permission; re-examine

07 / What would count as a demonstration?

Positive case: a bounded permitted action executes only under a currently valid, locally enforceable authorization and produces an observable outcome. Negative cases: each invalid or unsupported action is demonstrably prevented from actuating. Evidence package: original observations, provenance, authority records, rule version, disposition, local enforcement trace, outcome, and unresolved deviations.

08 / Human intervention and changed context

Human presence is not human governance. An override must be examined for identity, legitimate scope, standing, timely causal reach to the governing point, actual state change, and effect on the consequence path. Effective intervention does not automatically establish admissible authority.

A new fire-alarm state, sensor-quality failure, network discontinuity, delegation change, or controller mode change may invalidate an earlier authorization. Commissioned once does not mean authorized forever.

09 / Image-by-image teaching specification

01 THE CONNECTED BUILDING

Visual: Distinct systems exchange data; no command arrow reaches equipment

Samantha teaching line: Connection is not permission.

02 THE PROPOSED CONSEQUENCE

Visual: Agent proposal card: target, duration, intended effect, uncertainty

Samantha teaching line: A proposal is not authorization.

03 REALITY + RECORD

Visual: Raw observations, source identity, timestamps, occupancy context

Samantha teaching line: Preserve what was observed.

04 CONTINUITY + ADMISSIBILITY

Visual: Visible trace with stale data, gap and mapping challenge

Samantha teaching line: A broken bridge cannot be assumed.

05 AUTHORITY + STANDING

Visual: Owner delegation, scope, revocation and specific asset binding

Samantha teaching line: Authority must fit this action now.

06 LOCAL EXECUTION BOUNDARY

Visual: Target controller, active mode, interlocks and allow/block gate

Samantha teaching line: Execution authority is established locally.

07 FOUR DISPOSITIONS

Visual: Four separate evidence states, four bounded outcomes

Samantha teaching line: ALLOW / HOLD / DENY / ESCALATE are not interchangeable.

08 OUTCOME + NEW REALITY

Visual: Enforced action or block, measured results, revalidation loop

Samantha teaching line: Observe the outcome; preserve the adverse record.

10 / Showroom production constraints

One distinct instructional image per scene. Use clear numbered teaching labels and physically plausible controls. Do not invent vendor endorsements, logos, or portraits. Do not merge independent governance architectures into one unexplained diagram. Samantha narrates the numbered elements outside the image. Keep image and narration independently navigable; no automatic scrolling.

11 / Publication and institutional independence

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12 / Source and invitation

Companion article: [The Future of Integration: Who Authorizes the Consequence?](#) (AutomatedBuildings.com, October 2026).

TA14 Exchange: www.ta14exchange.com. Invite practitioners to submit one bounded integration-to-action scenario and the evidence and local authority they believe make it executable. The purpose is examination, not predetermined agreement.